Math 105L **Lecture 5-1**

Limits, Continuity, & One-Sided Limits

1. The Three Conditions for Continuity at $x = c$

A function $f(x)$ is continuous at a point $x = c$ if and only if:

1. $\mathbf{f(c)}$ is defined (the function exists at the point).
2. $\lim_{\mathbf{x \to c}} \mathbf{f(x)}$ exists (the values approach a single finite real number).
3. $\lim_{\mathbf{x \to c}} \mathbf{f(x)} = \mathbf{f(c)}$ (what we expect the function to be matches what it actually is).

Three Ways Continuity Can Fail (Types of Discontinuities)

- **1. Hole / Removable Discontinuity:** We know what to expect as $x \to c$, but there is a hole in the graph ($f(c)$ is undefined or detached from the curve).
- **2. Infinite Discontinuity (Vertical Asymptote):** The values grow without bound ($f(x) \to \pm\infty$ as $x \to c$).
- **3. Jump Discontinuity:** The function approaches two completely different finite values from the left and from the right ($\lim_{x \to c^-} f(x) \neq \lim_{x \to c^+} f(x)$).

2. One-Sided Limits & Two-Sided Limit Existence

The Fundamental Criterion for Limit Existence:

$\lim_{x \to c} f(x) = L$ exists if and only if both one-sided limits exist and are equal to L:

$$\lim_{\mathbf{x \to c}} \mathbf{f(x)} = \mathbf{L} \iff \lim_{\mathbf{x \to c^+}} \mathbf{f(x)} = \mathbf{L} = \lim_{\mathbf{x \to c^-}} \mathbf{f(x)}$$

- $\lim_{\mathbf{x \to c^+}} \mathbf{f(x)}$: Right-hand limit (approaching c from numbers greater than c).
- $\lim_{\mathbf{x \to c^-}} \mathbf{f(x)}$: Left-hand limit (approaching c from numbers less than c).

3. Worked Classroom Analysis: Piecewise & Rational Limits

- **Example 1: The Signum Function $f(x) = \frac{|x|}{x}$ at $x = 0$:** Using the piecewise definition $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$:

$$f(x) = \begin{cases} \frac{x}{x} = 1, & x > 0 \\ \frac{-x}{x} = -1, & x < 0 \end{cases}$$

  - Right-hand limit: $\lim_{x \to 0^+} \frac{|x|}{x} = \lim_{x \to 0^+} (1) = \mathbf{1}$.
  - Left-hand limit: $\lim_{x \to 0^-} \frac{|x|}{x} = \lim_{x \to 0^-} (-1) = \mathbf{-1}$.
  - Since $1 \neq -1$, the two-sided limit $\lim_{\mathbf{x \to 0}} \frac{\mathbf{|x|}}{\mathbf{x}}$ Does Not Exist (DNE) (Jump Discontinuity).

- **Example 2: The Reciprocal Function $f(x) = \frac{1}{x}$ at $x = 0$:**
  - As $x \to 0^+$, denominators are positive and shrink to 0: $\lim_{x \to 0^+} \frac{1}{x} = +\infty$.
  - As $x \to 0^-$, denominators are negative and shrink to 0: $\lim_{x \to 0^-} \frac{1}{x} = -\infty$.
  - The two-sided limit $\lim_{\mathbf{x \to 0}} \frac{\mathbf{1}}{\mathbf{x}}$ DNE (Infinite Discontinuity / Vertical Asymptote).